Machine Learning

- IID (identically, independently distributed)
- two main problem

$$\begin{cases} \text{Estimation} & : \text{training} \\ \text{Generalization} & : \text{test} \end{cases}$$

Supervised Learning

(i) classification $\begin{cases} \text{binary} \\ \text{multi-class} \end{cases}$

(ii) regression : $\mathbb{R}^d \longrightarrow \mathbb{R}^k$

Unsupervised Learning

(i) density estimation

: predict $P(x^{(n+1)})$

(ii) clustering

(iii) dimensionality reduction

Reinforcement Learning

상태 $x^{(0)}$ → 행동 $y^{(0)}$ → 보상 $r^{(0)}$ → 상태 $x^{(1)}$

Sequence Learning, $x_0 \cdots x_n \longrightarrow y_0 \cdots y_n$

(i) f : compute next hidden state.

(ii) g : compute output given current hidden state.

- Semi-Supervised : both $(x^{(i)}, y^{(i)})$ and $x^{(i)}$ only
- Active Learning : select $x^{(i)}$ and ask (high cost) for $y^{(i)}$
- Transfer Learning (Meta-Learning)

Evaluation Criteria

- 0-1 Loss $\begin{cases} 0 & g = a \\ 1 & \text{otherwise} \end{cases}$

- squared Loss $(g-a)^2$

- Linear Loss $|g-a|$

- Asymmetric Loss $\begin{cases} 1 & g=1 \quad a=0 \\ 10 & g=0 \quad a=1 \\ 0 & \end{cases}$

Optimization

- prediction rule / hypothesis $h(x;\theta)$

- model class $\mathcal{M}$

  $\Rightarrow$ parameterized by vector of params Θ

- $h(x;\theta,\theta_0) = \theta^T x + \theta_0$

- training error $\mathcal{E}_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} L(h(x^{(i)};\theta), y^{(i)})$